

Joint Stiffness–Damping Theory of the Non-Normal Grid

First theoretical results on the pair $(\tilde{L}, \tilde{D})$, each verified numerically

What is new here

The standard spectral theory of low-inertia grids keeps inertia as a geometry but collapses damping to a scalar $\tilde{D} \approx \gamma I$. We keep both: stiffness $\tilde{L} = M^{-1/2} L M^{-1/2}$ and damping $\tilde{D} = M^{-1/2} D M^{-1/2}$ as two operators. The central object becomes the quadratic pencil $(s^2 I + s \tilde{D} + \tilde{L})$. We prove three results, each checked on a worked 3-bus case: (1) heterogeneous damping couples the modes even with no PLL; (2) the right measure of this coupling is the off-diagonal of $\tilde{D}$ in the eigenbasis of $\tilde{L}$, not the departure-from-normality of the lifted operator; (3) where added damping helps is node-dependent, and the ranking is not the inertia ranking.

1 Setup: the quadratic pencil

The mass-normalized relative dynamics are

$$\ddot{\vartheta} + \tilde{D} \dot{\vartheta} + \tilde{L} \vartheta = M^{-1/2} \tilde{p}, \quad \tilde{L} = \tilde{L}^\top \succeq 0, \quad \tilde{D} = \tilde{D}^\top \succeq 0. \quad (1)$$

The poles are the roots of the *quadratic eigenvalue problem* (QEP)

$$(s^2 I + s \tilde{D} + \tilde{L})v = 0. \quad (2)$$

Let $\tilde{L} q_k = \nu_k q_k$ with orthonormal $\{q_k\}$, $0 = \nu_1 < \nu_2 \leq \dots$, and $Q = [q_1, \dots, q_n]$.

Definition 1 (Classical vs. non-classical damping). Damping is *classical* (proportional) if $\tilde{D}$ is diagonal in the eigenbasis of $\tilde{L}$, i.e. $Q^\top \tilde{D} Q$ is diagonal. Otherwise it is *non-classical*.

2 Result 1: heterogeneous damping couples the modes (no PLL needed)

Theorem 1 (Modal coupling from damping)

Project (1) onto the eigenbasis of $\tilde{L}$ with $\eta = Q^\top \vartheta$. The modal dynamics are

$$\ddot{\eta} + \underbrace{(Q^\top \tilde{D} Q)}_{\hat{D}} \dot{\eta} + \text{diag}(\nu_k) \eta = Q^\top M^{-1/2} \tilde{p}. \quad (3)$$

If $\tilde{D}$ is classical, $\hat{D}$ is diagonal and the modes are independent oscillators (the standard scalar theory). If $\tilde{D}$ is non-classical, $\hat{D}$ has nonzero off-diagonal entries $\hat{D}_{k\ell} = q_k^\top \tilde{D} q_\ell \neq 0$, and mode k is driven by the velocity of mode ℓ . The closed-form pole expression $s_{k,\pm} = -\frac{\delta_k}{2} \pm \sqrt{(\frac{\delta_k}{2})^2 - \nu_k}$ is then *invalid*: the poles no longer separate by mode.

Proof. Substitute $\vartheta = Q\eta$ in (1) and left-multiply by $Q^\top$; use $Q^\top \tilde{L} Q = \text{diag}(\nu_k)$ (orthonormality) and $Q^\top \tilde{D} Q = \hat{D}$. The off-diagonal of $\hat{D}$ measures exactly the failure of (2) to decouple. $\square$

Verified (3-bus): the coupling is real

With $\tilde{L}$ from the 3-bus case and $\tilde{D}$ from $D = \text{diag}(0.4, 1.2, 0.5)$ (non-classical):

$$\left\| \text{off-diag}(\hat{D}) \right\|_F = 0.246 \neq 0, \quad \text{vs.} \quad \tilde{D} = cI \Rightarrow \left\| \text{off-diag}(\hat{D}) \right\|_F = 0.000.$$

The classical scalar theory is the special case where this number is zero. Heterogeneous control damping makes it nonzero *before any PLL is added*.

3 Result 2: the correct measure of the second non-normality source

A trap we avoid. The lifted operator $A_{\text{sys}} = \begin{bmatrix} 0 & I \\ -\tilde{L} & -\tilde{D} \end{bmatrix}$ is *always* non-normal, even with $\tilde{D} = cI$: in the 3-bus case $d_F(A_{\text{sys}}) = 4.22$ regardless. So the departure-from-normality of A_{sys} does *not* isolate the damping effect. One must measure the coupling intrinsically.

Definition 2 (Stiffness–damping misalignment index).

$$\mathcal{C}(\tilde{L}, \tilde{D}) := \left\| Q^\top \tilde{D} Q - \text{diag}(Q^\top \tilde{D} Q) \right\|_F \quad (4)$$

the Frobenius norm of the off-diagonal of $\tilde{D}$ in the eigenbasis of $\tilde{L}$. Equivalently, $\mathcal{C} = 0$ iff $\tilde{L}$ and $\tilde{D}$ commute, $[\tilde{L}, \tilde{D}] = 0$.

Proposition 1 (Commutator form). $\mathcal{C}(\tilde{L}, \tilde{D}) = 0 \iff \tilde{L}\tilde{D} = \tilde{D}\tilde{L}$. The two geometries share an eigenbasis exactly when they commute; the commutator $\left\| [\tilde{L}, \tilde{D}] \right\|_F$ is a basis-free certificate of misalignment.

Verified (3-bus)

$\left\| [\tilde{L}, \tilde{D}_{\text{classical}}] \right\|_F = 0.000$ and $\left\| [\tilde{L}, \tilde{D}_{\text{hetero}}] \right\|_F = 1.223$. The misalignment index $\mathcal{C}$ and the commutator agree on which case decouples and which does not.

4 Result 3: where added damping helps is node-dependent

In classical theory, adding damping anywhere helps every mode. With non-classical damping the effect is routed by the coupling $\hat{D}$, so the benefit of damping at a given node depends on that node's role in the coupled modes.

Theorem 2 (Damping placement is not the inertia ranking)

Let $g_{\max} = \sup_{\Omega} \sigma_{\max}(C(i\Omega I - A_{\text{sys}})^{-1}B)$ be the worst-case transient amplification. The sensitivity of $g_{\max}$ to damping added at node i , $\partial g_{\max} / \partial D_i$, is in general not ordered by $1/M_i$ nor by the modal participation $q_{c,i}^2$. The most effective node for damping can differ from the most fragile node for inertia.

Verified (3-bus): damping ranking $\neq$ inertia ranking

Base case (hetero damping + PLL), peak $g_{\max} = 5.75$. Adding +0.5 damping per node:

add damping at	N1 (SG)	N2 (GFL)	N3 (GFM)
$\Delta g_{\max}$	-1.52	-0.29	-2.08

Damping at N3 helps most (-2.08); damping at N2, the low-inertia GFL node one might expect to target, helps least (-0.29). The transient is best damped at a node other than the weakest-inertia one. The classical intuition (damp the lightest node) gives the wrong answer.

5 Synthesis: two sources, one certificate

Non-normality of the grid operator has two distinct origins, now both quantified:

1. **Stiffness–damping misalignment** $\mathcal{C}(\tilde{L}, \tilde{D}) > 0$: heterogeneous control damping that does not commute with the network stiffness. Present with no PLL.
2. **Control asymmetry** from the grid-following PLL: the one-sided frequency-to-angle gain in A_{sys} .

Both feed the same diagnostic, the resolvent peak $g_{\max}$, and both are invisible to a spectrum-only (ν_c) reading. The first is the genuinely new piece of this development: prior spectral hosting theory sets $\tilde{D} \approx \gamma I$ and therefore has $\mathcal{C} \equiv 0$ by assumption.

The first strong

Dynamic fragility in inverter grids is governed by the joint geometry of stiffness and damping, not by stiffness alone. The misalignment $\mathcal{C}(\tilde{L}, \tilde{D}) = \left\| [\tilde{L}, \tilde{D}] \right\|$ is a new, basis-free fragility source that the mass-normalized Laplacian ν_c cannot see, and it reorders where damping should be placed.

Verified, not asserted. Every boxed number above was computed on the 3-bus case ($\tilde{L}$ from $w = (5, 4, 3)$, $M = \text{diag}(10, 2, 6)$, $D = \text{diag}(0.4, 1.2, 0.5)$, PLL gain $\kappa_p = 1$). The next step is to measure $\mathcal{C}(\tilde{L}, \tilde{D})$ and the damping-placement ranking on the IEEE 39-bus system and compare against the inertia and SCR rankings.